

AI Agents for Materials Discovery

Retrieval-augmented, property-aware and generative machine learning for inorganic crystal design

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Format note. Drafted to the MSci brief: ≈ 8 pages of literature review (§1–§7) and ≈ 2 pages of research proposal (§8–§12), ≤ 10 pages total.

Abstract

Large language models (LLMs) have made scientific knowledge conversationally accessible, and retrieval-augmented generation (RAG) grounds their answers in trusted sources. Yet LLMs remain unreliable on the *continuous, numerical* reasoning that materials science demands, and pure RAG can *retrieve and summarise* facts but cannot *compute* a property or *propose* a new compound. In parallel, two specialised model families have matured: graph neural networks (GNNs) that predict material properties at a fraction of the cost of density functional theory (DFT), and diffusion-based generative models that produce stable, property-conditioned crystals. This review surveys foundation models, RAG/agent frameworks, property-prediction networks and generative models for inorganic materials, and argues that these strands remain largely *siloed*. The proposed project closes that gap by extending a materials RAG framework into a tool-using *LLM agent* that delegates numerical property prediction to dedicated ML models and crystal generation to a diffusion model, turning a literature-retrieval assistant into an end-to-end, query-driven discovery engine.

1. Introduction and motivation

Materials science is entering a phase of data-driven discovery. Computational materials research has long relied on first-principles methods such as density functional theory (DFT) and molecular dynamics, which are physically principled but expensive: a single DFT relaxation can take hours of CPU time, making exhaustive search of chemical space infeasible.¹ Machine learning trained on computational and experimental data now supplements these simulations, predicting properties and generating designs at a fraction of the cost.^{1,19} Materials data is, however, demanding: it is multimodal (structures, text, spectra, images), spans electronic to macroscopic scales, must obey physical laws such as energy conservation and symmetry, and is scarce relative to the corpora that train language models.¹ Such models have also been narrow, each property or modality requiring a bespoke pipeline.

The arrival of **foundation models** (FMs), large models pretrained on broad data that transfer across many downstream tasks, has reframed the field.³⁵ Following the impact of the Transformer⁴ and its descendants BERT⁵ and GPT⁶ in natural-language processing, the materials community now asks whether similarly general models can unify reasoning over atomic structures, chemical formulae, spectra and the text of the literature.¹ Two capabilities have proved especially consequential. First, **retrieval-augmented generation** (RAG) couples an LLM to an external knowledge store so that responses are grounded in retrieved evidence rather than parametric memory alone, mitigating hallucination.⁷ Second, **LLM agents** wrap a language model in a reasoning loop that can call external tools, query databases and act on the results, allowing the model to outsource tasks it cannot perform reliably on its own.³⁰

This last point is the crux of the project. LLMs are powerful interfaces to *qualitative* knowledge but are demonstrably weak at the *quantitative*, continuous reasoning at the heart of materials science: predicting a band gap, a formation energy or an elastic modulus to useful accuracy.³¹ A RAG system can find that a compound's band gap was reported as 1.6 eV; it cannot, by itself, predict the band gap of a compound that has never been measured, nor can it generate a novel crystal that should exhibit a target band gap. Those are precisely the tasks at which specialised ML such as GNN property predictors^{19,20} and diffusion-based crystal generators^{12,13} now excels. This review therefore traces a deliberate arc: from the general

landscape of AI for materials, through RAG and agents, to property prediction and generative modelling, ending at the integration gap that the project will address.

The review is scoped to **inorganic crystalline materials** and to the four capabilities the project must combine: knowledge retrieval, agentic orchestration, property prediction, and property-conditioned generation. Following the assignment's guidance it foregrounds key ideas and representative methods rather than exhaustive coverage; a recent survey of AI for materials science¹ provides the backbone, supplemented throughout with the primary literature.

2. Foundation models and the AI-for-materials task landscape

It is useful to begin with a map of the problems that AI is being applied to in materials science, because the project sits at the intersection of several of them. Van *et al.*¹ organise the field into six recurring task families (Table 1); the project draws principally on T1, T3 and T4, orchestrated by an agent.

Table 1. The six task families that recur across AI-for-materials research, after the taxonomy of Van *et al.*¹ Tasks central to this project are highlighted.

Task	Family	What it involves	Project
T1	Data extraction, interpretation & Q&A	Mining facts and relations from papers/patents; question answering over the literature, the natural home of RAG.	<i>core</i>
T2	Atomistic simulation	Machine-learned interatomic potentials replacing/accelerating DFT and MD.	context
T3	Property prediction	Mapping composition/structure to properties (band gap, formation energy, moduli).	<i>core</i>
T4	Structure, design & discovery	Generating novel, stable, property-targeted structures (inverse design).	<i>core</i>
T5	Process planning & optimisation	Synthesis route planning, experimental optimisation.	extension
T6	Multiscale modelling	Bridging electronic, atomistic and continuum scales.	out of scope

Cutting across these tasks, FMs are conventionally grouped into three categories.¹ **Unimodal** models operate within a single modality: text (e.g. the BERT-derived MatSciBERT²³) or structure (GNNs over crystal graphs). **Multimodal** models attempt to fuse two or more modalities, for example aligning text with molecular structure, but remain comparatively immature: most deployed models still reason in a single modality.¹ **LLM agents** extend a base model with external knowledge and tools, and are the category the project lives in. Architecturally, the three Transformer variants recur throughout: *encoder-only* models (BERT-style) build representations well-suited to property prediction; *decoder-only* models (GPT-style) generate sequences and underpin string-based crystal generation; and *encoder-decoder* models support translation-like tasks.^{4,5}

The strategic reading of this landscape is that *no single model does everything well*. Text models read the literature but cannot relax a structure; potentials and GNNs compute properties but cannot converse; generators propose structures but neither explain their reasoning nor guarantee stability. This division of labour is exactly why an *agentic* architecture, a language model that routes sub-problems to specialised tools, is the natural unifying abstraction, and why it forms the spine of the proposed system.

3. Retrieval-augmented generation and LLM agents for materials

3.1 Why retrieval, and why grounding matters

LLMs hold much materials knowledge in their parameters, but it is static, opaque and prone to *hallucination* outside the training distribution.¹ RAG retrieves relevant documents at query time and conditions generation on them, so claims are traceable to sources;⁷ in a domain where an invented property value is worse than no answer, this grounding is a requirement, not a convenience. Grounding does not, however, fix numerical capability. Miret and Krishnan find general-purpose LLMs are not yet reliable for real-world materials discovery, especially on quantitative tasks,³¹ and on LLM4Mat-Bench LLMs asked to regress properties sit near a mean-predictor baseline (MAD:MAE about 1) while a task-specific GNN such as

CGCNN scores several-fold higher (around 5.3 on Materials Project).^{43,38} This is the project's central premise: numerical work should be *delegated* to specialised ML, not learned in context.

3.2 RAG systems for materials knowledge

The clearest exemplar is **HoneyComb**,²⁴ an LLM agent built around three components: *MatSciKB*, a curated knowledge base aggregating arXiv articles, Wikipedia, datasets and textbooks; *ToolHub*, a suite of retrieval tools for up-to-date information; and a *Retriever* that pulls relevant knowledge using classical and neural retrieval (BM25 and Contriever). HoneyComb is model-agnostic and measurably improves domain reasoning over a bare LLM, a template for the retrieval layer of the proposed system. Its predecessor HoneyBee²⁵ showed that instruction-tuning an open LLM on materials text (with one model generating instructions and another verifying them) already lifts domain competence, establishing the data-curation pattern the project can reuse.

3.3 From retrieval to action: tool-using agents

The decisive step beyond retrieval is *tool use*. **ChemCrow**³⁰ equips an LLM with eighteen chemistry tools and a reason-then-act loop (the ReAct pattern³⁹) in LangChain, performing tasks no bare LLM can. Most relevant is **LLMatDesign**,²⁶ almost a blueprint for this project: from a target composition and property it proposes an elemental modification, calls ML tools (MatDeepLearn, TorchMD-Net³²) to predict the result, evaluates it against the target, and iterates via Materials Project API calls. **ChatMOF**²⁷ applies the same agent, toolkit and evaluator pattern to metal-organic frameworks, spanning retrieval, prediction and generation, while human-in-the-loop systems (**MatAgent**,²⁸ **MatPilot**²⁹) add expert oversight and, for MatPilot, robotic execution.

ChatMOF and **LLMatDesign** partly pre-empt the present design: ChatMOF²⁷ already unifies retrieval, prediction and generation, and LLMatDesign²⁶ already runs the predict-evaluate-iterate loop the project reuses, so the architectural pattern is not new. What this project adds is the particular instantiation that neither provides together: a diffusion generator for *general inorganic crystals* (rather than MOFs or genetic-algorithm modification), literature-grounded RAG with explicit faithfulness evaluation, and a verification and abstention layer that gates every returned candidate.

Two points follow. These agents show the value of delegating numerical work to specialised models, yet retrieval and generation are rarely unified: HoneyComb retrieves but does not generate crystals, while the diffusion generators of §5 generate but are not wrapped in a grounded, conversational interface.

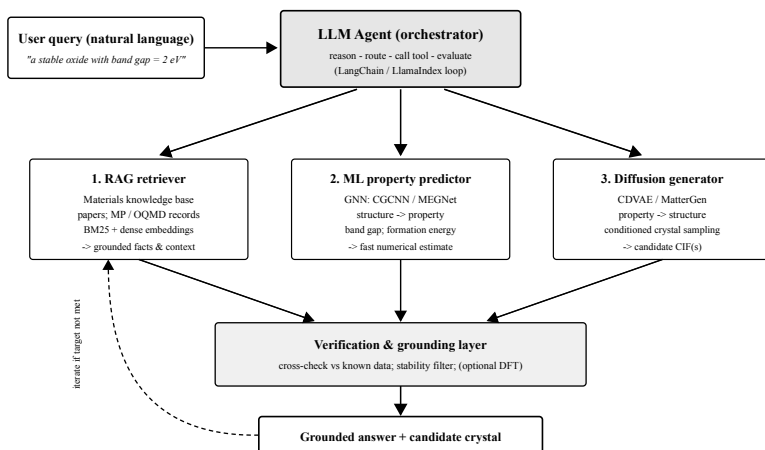


Figure 1. Proposed agentic architecture (original schematic). A single LLM agent interprets a natural-language query and routes sub-tasks to three specialised tools: ① a RAG retriever for grounded knowledge, ② an ML property predictor for fast numerical estimates, and ③ a diffusion model for property-conditioned crystal generation, then verifies candidates against known data before answering, iterating when the target property is unmet.

4. Machine learning for material property prediction

Property prediction is the quantitative engine the agent will call, and the subfield where ML most clearly outperforms an LLM. The premise is simple: DFT is accurate but expensive, so learn the structure→property mapping directly from data and predict in milliseconds.¹ Two model families dominate, distinguished mainly by how they represent a material.^{19,20}

4.1 Graph neural networks over structures

Crystals and molecules are naturally graphs (atoms as nodes, bonds or proximity as edges), and *geometric* GNNs exploit this. **SchNet**²¹ introduced continuous-filter convolutions for rotation-invariant quantum interactions; **CGCNN**¹⁹ represents a periodic crystal as a multi-edge graph and is notably *interpretable*, attributing global properties to local environments; and **MEGNet**²⁰ generalises this with atomic, bond and global-state attributes, predicting formation energies, band gaps and elastic moduli across broad chemical space when trained on Materials Project² data. On formation energy the two are close but separable (MAE about 0.034 eV/atom for CGCNN versus 0.025 for MEGNet under the controlled Matbench splits³⁷), so the project adopts **MEGNet** as its default predictor and keeps CGCNN as an interpretable cross-check. A related family, machine-learned interatomic potentials such as MACE,¹¹ MatterSim¹⁰ and CHGNet,⁴¹ approximates DFT energies and forces and supplies the fast energy-above-hull estimates the project reuses for stability checking (§9).

4.2 Transformer- and string-based predictors

An alternative encodes a material as a string for a Transformer: **ChemBERTa**²² adapts RoBERTa to molecular SMILES, with related encoder-only models for catalysis and solvation. Because SMILES and SELFIES are 2D, crystal-aware variants (Matformer, ComFormer) inject lattice geometry and periodicity into the attention mechanism.¹ The representation question (2D strings versus 3D graphs) recurs: 3D structure yields richer features for spatially dependent properties, which is why GNNs remain the default for inorganic crystals.

Crucially for the project, property predictors are also what make *generation* useful: a generator proposes a candidate, and a predictor (or DFT) scores whether it actually meets the target. Standardised benchmarks such as MoleculeNet³³ and Matbench³⁷ allow these predictors to be compared and selected on like-for-like tasks, which the project will use to choose and validate its predictor.

5. Generative and diffusion models for materials

If property prediction asks "what is the property of this material?", generative modelling asks the inverse and harder question: "what material has this property?" Early work used VAEs and GANs, but both are limited for crystals (GANs suffer mode collapse, VAEs give lower-fidelity samples, and neither handles periodic symmetry natively). **Diffusion models** displaced them through higher sample quality, stable training and an *equivariant* denoiser that respects crystal symmetry, jointly denoising lattice, fractional coordinates and atom types.⁸ Property conditioning then steers sampling toward a target, though the mechanism is model-specific. One distinction matters throughout: *structure prediction* fixes the composition, whereas *de-novo generation* leaves it free; the project needs the latter, to propose novel compositions.^{14,12}

5.1 Graph- and diffusion-based crystal generation

G-SchNet¹⁸ pioneered equivariant 3D molecule generation. **CDVAE**¹³ was a landmark for inorganic solids, pairing periodic GNN encoder and decoder with a diffusion process to generate *stable* crystals and demonstrating property optimisation; its conditional extension **Con-CDVAE**¹⁶ samples latent variables from a target property (e.g. formation energy or band gap) and decodes a structure, the exact capability the project envisages. **DiffCSP**¹⁴ instead targets crystal *structure prediction* at fixed composition, jointly diffusing lattice and fractional coordinates, with an extension that also diffuses atom types; **DiffCSP++**¹⁵ adds space-group constraints.

5.2 Scaling up: MatterGen, GNoME and language-model generators

The current flagship is **MatterGen**,¹² a diffusion model that generates stable, diverse inorganic crystals and, decisively, is fine-tuned with property labels to steer generation under constraints; it reports structures more than twice as likely to be both novel and stable as prior models such as CDVAE and far closer to the DFT minimum,¹² which is why the project takes it as the baseline generator with CDVAE/Con-CDVAE as fallback (§11). A complementary, search-driven approach is **GNoME**,⁹ a *screening* pipeline that enumerates and filters candidates with GNN energy ensembles and DFT validation; it reported about 2.2 million structures below the convex hull, of which about 381,000 were predicted stable, though these figures have since been critiqued.⁴⁷ A third route treats crystals as text: **CrystaLLM**¹⁷ generates CIF files by next-token prediction, and fine-tuned general LLMs can likewise emit valid crystals as text,⁴² a representation well suited to an LLM-centric agent.

5.3 Conditioning a generator on a target property

The three candidate generators differ in *how* they impose a target property, which has direct practical consequences. **CDVAE**¹³ conditions only indirectly, by gradient-ascent search in its latent space against a separately trained property predictor; **Con-CDVAE**¹⁶ instead learns a property-conditioned prior so a target value is supplied at sampling time; and **MatterGen**¹² fine-tunes the base diffusion model with a property adapter and applies guidance during denoising. The project commits to MatterGen as the baseline generator because it ships ready property-conditioned checkpoints for band gap, bulk modulus, magnetic density and chemical system, removing the need to train conditioning from scratch, with Con-CDVAE and CDVAE retained as fall-backs should those checkpoints prove unsuitable.

RECURRING LIMITATION

Generative models produce *syntactically valid* structures that are not guaranteed to be *synthesisable* or even physically plausible; MatterGen itself cannot guarantee experimental feasibility, and generated structures can be physically implausible despite passing validity checks.^{12,1} This is exactly why the project pairs generation with a property-predictor/verification stage rather than trusting the generator's output directly; synthesizability itself can ultimately only be confirmed experimentally, as autonomous laboratories such as A-Lab³⁴ now begin to do at scale.

6. Datasets, tools and infrastructure

Every model above is only as good as its data. The computational backbone is a set of DFT databases, the **Materials Project**² and **OQMD**³ alongside NOMAD, JARVIS, the Open Catalyst sets and QM9, supplying hundreds of thousands of structures with computed properties for training and retrieval.¹ A pivotal caveat is **data imbalance**:¹ coverage skews to stable inorganic oxides at the atomic scale, while polymers, disordered phases and larger length scales are sparse. This bounds the project's claims: a system grounded on Materials-Project-like data is most trustworthy for stable inorganic crystals, which appropriately scopes the work.

Mature open-source tools make the project tractable within a year: *pymatgen*³⁶, ASE and the Materials Project / OPTIMADE APIs for structures and data; matminer and RDKit for featurisation; PyTorch Geometric and DGL for the GNNs; and LangChain and LlamaIndex for the RAG and agent scaffolding.¹ Reference implementations exist for the key models (CGCNN, MEGNet, CDVAE, MatterGen), so the project adapts proven components rather than rebuilding them, a deliberate de-risking choice (§11).

7. Synthesis: the integration gap and the case for an agent

Three conclusions emerge from the literature. **First**, LLMs are excellent linguistic interfaces but unreliable quantitative reasoners; grounding via RAG fixes *factual* reliability but not *numerical* capability.^{31,7} **Second**, dedicated ML now performs the two quantitative tasks the project needs, property prediction (CGCNN, MEGNet)^{19,20} and property-conditioned crystal generation (CDVAE, Con-CDVAE, MatterGen)^{13,16,12}, to a useful standard. **Third**, agent frameworks already show that an LLM can usefully delegate to such tools, and the most integrated of these go beyond a single capability: ChatMOF²⁷ already spans retrieval, prediction *and* generation, albeit for metal-organic frameworks, while several recent systems target inorganic crystals, including **GenMS**⁴⁴, **MatExpert**⁴⁵ and a diffusion-based **MatAgent**⁴⁶. The surviving distinction is therefore not whether an agent can close the loop at all, but how: existing systems variously restrict the chemistry (MOFs rather than general inorganic crystals), the generator (genetic-algorithm or string-based rather than diffusion) or the grounding (tabular or knowledge-base lookup rather than literature-grounded RAG with explicit faithfulness evaluation), and few couple all three to an explicit verification and abstention layer.

THE RESEARCH GAP

Materials RAG systems *retrieve and summarise* but cannot *predict* unmeasured properties or *generate* new structures; property predictors and diffusion generators can do exactly this but are not typically exposed through a grounded, conversational interface. Integrated agents do exist: ChatMOF²⁷ closes the retrieve-predict-generate loop for metal-organic frameworks, and GenMS, MatExpert and a diffusion-based MatAgent target inorganic crystals. The contribution here is therefore **bounded rather than absolute**: the combination the project pursues, namely *literature-grounded RAG over scientific text with faithfulness evaluation* (rather than tabular or knowledge-base lookup), a *diffusion generator* (rather than a genetic-algorithm or string-based one), coverage of *general inorganic crystals* (rather than MOFs), and an *explicit verification and abstention layer* that gates every returned candidate, is not yet offered together by any single published system. Assembling and evaluating that specific combination is the project.

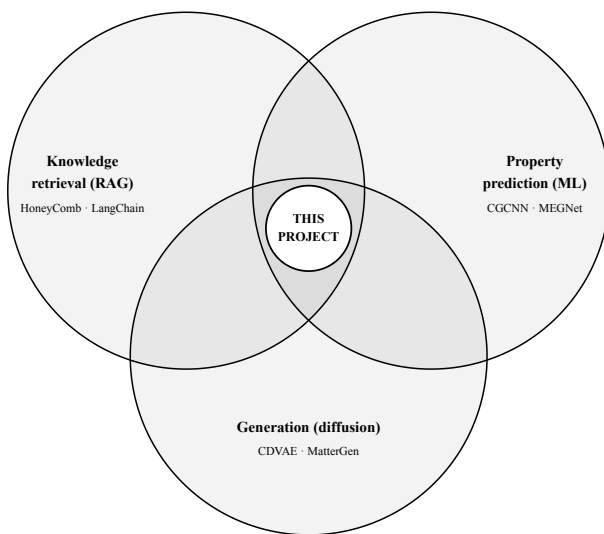


Figure 2. Positioning of the project (original schematic). Each capability is mature in isolation, and recent agents already pair them: ChatMOF²⁷ reaches the three-way intersection for MOFs, and GenMS, MatExpert and a diffusion-based MatAgent target inorganic crystals. What remains lightly occupied is the specific configuration the project adopts: literature-grounded retrieval *and* numerical prediction *and* diffusion-based, property-conditioned generation over general inorganic crystals, gated by an explicit verification and abstention layer in one agentic loop.

Table 3. Capability comparison of recent LLM-driven materials systems against the proposed agent. Cells: ✓ present, partial, ✗ absent.

System	Literature-grounded RAG	ML property prediction	Diffusion property-conditioned generation	Verify / abstain loop	Material class
ChatMOF ²⁷	partial	✓	✗	✗	MOFs
LLMatDesign ²⁶	✗	✓	✗	partial	inorganic crystals
GenMS ⁴⁴	✗	✓	partial	✗	inorganic crystals
MatAgent ⁴⁶	partial	✓	partial	✗	inorganic crystals
This project	✓	✓	✓	✓	inorganic crystals

An agentic architecture is the natural way to close it because the agent need not *be* good at chemistry; it needs to know *which tool* to call and how to compose the results, exactly the competence LLMs do possess.³⁰ The remainder of this document specifies how.

8. Research proposal: aims and objectives

Background in one paragraph. The project extends an existing materials-RAG framework into a tool-using LLM agent. Where current RAG systems retrieve and summarise materials knowledge, the agent will additionally (i) delegate *property prediction* to a dedicated ML model so it can answer quantitative questions about unmeasured materials, and (ii) delegate *query-based generation* to a diffusion model so it can propose novel crystals that should exhibit a requested property, each result grounded against retrieved data and verified before it is returned.

AIM

To design, build and evaluate an LLM-agent system that unifies retrieval-augmented materials knowledge, ML-based property prediction and diffusion-based crystal generation, enabling natural-language, property-targeted discovery of inorganic crystalline materials with grounded, verifiable outputs.

Objectives.

- O1. O1 RAG baseline.** Build the knowledge base from Materials Project and literature abstracts, implement hybrid *BM25* plus dense retrieval, and characterise its limits on quantitative queries. *Acceptance:* on the fixed query set of O5, retrieval reaches a recall@10 of at least 80% for entities present in the KB, with quantitative-query failure modes documented.
- O2. O2 Property predictor as a tool.** Wrap pretrained *CGCNN/MEGNet* to predict band gap and formation energy, exposed as a callable tool and benchmarked on *Matbench*. *Acceptance:* on a held-out split, the wrapped predictor's MAE is within $1.2\times$ of the published Matbench MEGNet formation-energy MAE of about 0.025 eV/atom³⁷, confirming the wrapper preserves baseline accuracy.
- O3. O3 Generative model as a tool.** Expose property-conditioned crystal generation by sampling from a pretrained *MatterGen* checkpoint (primary), with *CDVAE/Con-CDVAE* as fallback. *Acceptance:* a sampled batch yields structures that are **valid** (charge/structure checks via *pymatgen*) **and property-matched** to the target within 0.5 eV for band-gap targets, scored by the independent oracle of O5, not the in-loop GNN.
- O4. O4 Agent loop.** Implement routing, tool-calling, a verification stage, and an iterate-if-unmet controller that re-queries tools when targets are not satisfied. *Acceptance:* on the O5 query set the agent routes to the correct tool in at least 90% of cases and the verification stage triggers at least one corrective iteration on under-met targets.
- O5. O5 End-to-end evaluation.** Evaluate against a **fixed held-out benchmark of about 50 natural-language property-target queries with gold answers**, reporting retrieval faithfulness, prediction error versus DFT, generation validity/novelty/property-match, and hallucination rate, with LLM-only / retrieval-only / full ablations. *Acceptance:* hallucination rate is measured by an LLM-judge / RAGAS-style faithfulness protocol (the RAGAS faithfulness metric⁵⁰), and every property-match claim is scored by an **independent oracle** (DFT single-points or a held-out predictor), never the in-loop GNN; the full agent must beat both ablations on property-match and hallucination rate.

9. Methodology and system design

The system (Fig. 1) is an LLM *orchestrator* exposing three tools through a standard agent framework. **Retrieval (Tool ①):** a knowledge base of Materials Project records and literature abstracts is indexed with a hybrid of sparse (BM25) and dense passage retrieval,⁴⁰ following the HoneyComb pattern;²⁴ the agent cites retrieved passages so every factual claim is traceable. **Prediction (Tool ②):** a pretrained CGCNN/MEGNet^{19,20} maps a structure (or a retrieved/generated candidate) to a property estimate in milliseconds, supplying the numerical capability the LLM lacks. **Generation (Tool ③):** the primary generator is **MatterGen**,¹² sampled from a pretrained property-conditioned checkpoint to draw candidate crystals for a requested property, with *Con-CDVAE*¹⁶ and *CDVAE*¹³ as fall-backs. Conditioning is limited to properties for which a pretrained checkpoint exists (band gap, bulk modulus, magnetic density, chemical system); targets outside that set require either a fall-back generator or the compositional-modification loop (§9). Outputs are *Structure* objects handled with *pymatgen*.³⁶

The **agent loop** ties these together: the LLM parses the query, decides whether it needs a fact ($\rightarrow$ ①), a number ($\rightarrow$ ②) or a new structure ($\rightarrow$ ③, then ② to score it), composes the results, and passes candidates through a **verification layer** that cross-checks predicted properties against retrieved/known values and applies a stability filter (using a machine-learned potential such as CHGNet⁴¹ to estimate the energy above the convex hull), optionally invoking a single DFT relaxation for the top candidate. If a generated material misses the target property, the agent iterates, re-prompting the generator or proposing a compositional modification in the manner of LLMatDesign.²⁶ Development is incremental and tool-by-tool so that each component is independently testable, and reuses reference implementations rather than training large models from scratch, feasible within the GPU budget of a single MSci project, in contrast to the tens of thousands of GPU-hours that flagship models such as GNoME and MatterSim required.¹

The tools share a single **representation contract** so that outputs of one are valid inputs to the next. Retrieval returns text passages plus structured records (typically a chemical formula and, where available, a Materials Project identifier); a formula or identifier is resolved to a *pymatgen Structure*³⁶, which is converted to the multi-edge crystal graph consumed by the predictor, which in turn returns a scalar property estimate. The binding constraint is that structure-based GNNs such as CGCNN and MEGNet^{19,20} require a full 3D structure, not a bare composition: for a compound whose structure is not already in Materials Project, the structure must first be obtained, either by lookup or, for genuinely novel candidates, by generating it (a *generate-then-predict* ordering, Tool ③ then Tool ②). A composition-only predictor (Roost⁴⁸ or CrabNet⁴⁹) is retained as a fall-back when only a formula is available and no structure can be resolved or generated. Finally, the **verification layer** reconciles the two information sources: when a retrieved literature or DFT value exists for a candidate, it is treated as ground truth and the model prediction is reported alongside it with the discrepancy; the predictor is relied upon as the answer only where no measured value is retrievable, keeping numerical claims either grounded or explicitly flagged as estimates.

10. Programme of work

The plan spans the MSci project year (Oct 2026 – Jun 2027), front-loading literature consolidation and the lowest-risk components and deferring the most ambitious generation work, with explicit slack before the dissertation.

Work package	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun
WP1 · Literature & tooling set-up									
WP2 · RAG baseline (O1)									
WP3 · Property-predictor tool (O2)									
WP4 · Agent loop + verification (O4)									
WP5 · Generative tool (O3)									
WP6 · End-to-end evaluation (O5)									
WP7 · Dissertation & write-up									

Solid grey bars = core path; black hatching = higher-risk generation work; light hatching = ongoing work / slack. WP5 (generation) depends on WP4; the RAG + predictor system (O1, O2, O4) is a hard go/no-go milestone, demonstrable by end January. Objectives (O1–O5) map to §8.

11. Risk assessment, fall-backs and flexibility

The plan is deliberately modular so that value is delivered even if the most ambitious objective slips: a working retrieval-plus-prediction agent (O1–O2, O4) is already a complete, assessable outcome, with generation (O3) layered on top.

Table 2. Principal risks and mitigations.

Risk	Severity	Mitigation / fall-back
Property-conditioned diffusion generation (the headline-novel, install-fragile component): running a pretrained checkpoint is days of effort, but conditioning a generator on a new property is weeks of work plus a GPU budget, and training from scratch is out of scope	HIGH	Tier the work: the core path uses only a pretrained conditioned MatterGen/CDVAE checkpoint (days); conditioning on a new property is attempted only as a stretch goal within available GPU hours; training from scratch is explicitly avoided. Fall-backs if conditioning fails: compositional modification + re-prediction (LLMatDesign style) or string-based CrystalLM generation, both gated by the predictor and stability filter.
Generated structures unsynthesisable / physically implausible	MED	Treat generation as hypothesis-only ; gate every candidate through the predictor + stability filter + optional DFT; report property-match and validity rates honestly rather than claiming discovery.
LLM hallucination in the agent's reasoning/answers	HIGH	Mandatory citation of retrieved evidence; numerical answers must come from Tool ② (never the LLM); strict typed function-calling schemas with validation/retry. Fall-back: a deterministic rule-based router maps recognised intents (e.g. "band gap of X") straight to tools, bypassing LLM routing on core paths.
Property predictor inaccurate out-of-distribution	MED	Restrict scope to well-covered inorganic crystals; report predictor uncertainty; cross-validate against retrieved DFT values. Fall-back: the agent abstains (a calibrated "I don't know" using retrieval-coverage and uncertainty signals) rather than answering confidently outside its regime.
Data/compute access (GPU hours, API limits)	LOW	Rely on open data (Materials Project/OQMD) and reference checkpoints; cache retrieved records locally to respect rate limits; College HPC for any DFT. Fall-back: OPTIMADE federation / Alexandria if a single source is unavailable; nothing in the core path needs large-scale pretraining.
Grounding/hallucination metrics noisy or contested	LOW	Use multiple complementary measures and emphasise controlled ablation deltas (LLM-only vs RAG-only vs full), which remain informative even if absolute numbers are imperfect.
Integration/engineering overruns	MED	Tool-by-tool development with independent tests; the RAG + predictor milestone (Jan) is a self-contained deliverable; built-in slack (WP7) before the dissertation.

Flexibility. Decision points are explicit: if generation (WP5) is not yielding valid, property-matched candidates by the end of April, the project pivots to a deeper evaluation of the retrieval-plus-prediction agent (e.g. a broader property set, an ablation of retrieval strategies), which is itself a strong contribution and directly answers the project brief's primary objective.

12. Expected outcomes and deliverables

The project will deliver: (1) a working LLM-agent system integrating RAG, property prediction and (target) generation for inorganic crystals; (2) an evaluation comparing it against bare-LLM and retrieval-only baselines on faithfulness, prediction error, generation quality and hallucination rate; (3) a curated materials knowledge base and a reproducible codebase built on open tools; and (4) the MSci dissertation. The broader contribution is conceptual as much as engineering: a demonstration that the right division of labour (language model for orchestration and explanation, specialised ML for numbers and structures, retrieval for grounding) is a more reliable route to materials discovery than ever-larger general models alone.³¹

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